

The x/y swap in the CF4++ radial velocity grid

Notes for the I.C.A.R.A. 2026 presentation — HALO, Part 100 (9 October 2026)

What Cosmicflows-4 and CF4++ are

Cosmicflows-4 (CF4) is the catalogue: about 56,000 galaxies whose distances were measured independently of redshift (Tully-Fisher relation, Cepheids, supernovae and other methods). With both the distance and the observed velocity of each galaxy, its **peculiar velocity** can be derived, i.e. how much it moves with respect to the pure expansion of the Universe.

CF4++ is the next step (Courtois et al. 2025): the **reconstruction** of the local Universe from the catalogue. The galaxies are scattered points and their distances have large uncertainties, but gravity links density and velocity. Hélène Courtois's team therefore reconstructs a **continuous field** on a cube of 3D cells: the matter density (including dark matter) and the velocity (v_x , v_y , v_z) at every point, even where no galaxies are observed. The “++” marks the updated version of the first CF4 reconstruction.

The reconstruction does not give a single answer but **many realisations** compatible with the data. The distributed file, `CF4pp_mean_std_grids.npz`, contains for each cell the **mean** of the realisations and their **standard deviation** (the uncertainty): density, Cartesian velocities v_x , v_y , v_z and a pre-computed grid of the **radial velocity** v_r . It is from this velocity field that flow lines are followed down to the basins of attraction, revealing Laniakea and the Great Attractor.

What the radial velocity of a cell is

It is the part of the cell's velocity along the observer's line of sight: positive if the cell is receding, negative if it is approaching. It is obtained by projecting the velocity vector onto the direction of the cell:

$$v_r = \mathbf{v} \cdot \hat{\mathbf{r}} = (v_x \cdot x + v_y \cdot y + v_z \cdot z) / r$$

where (x, y, z) is the cell position and r its distance.

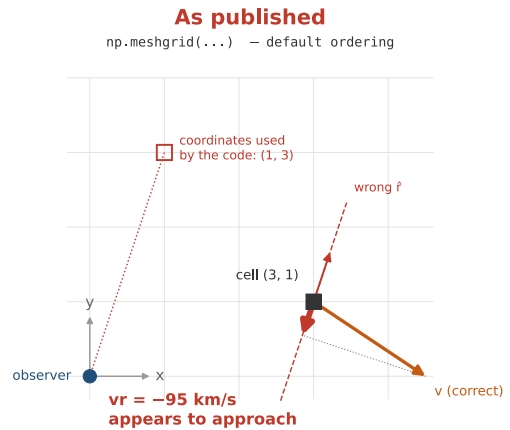
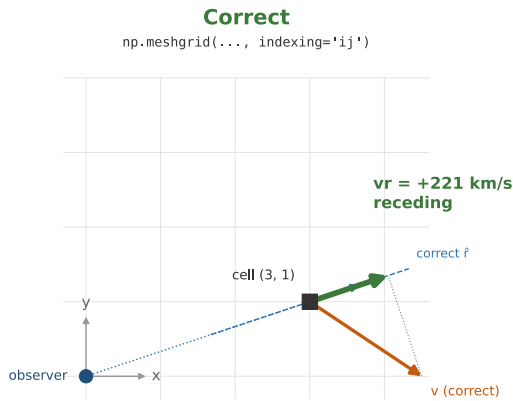
What went wrong

The coordinates (x, y, z) of every cell are usually built with the NumPy function `np.meshgrid`. With its default setting, however, `meshgrid` returns the first two coordinates **swapped** with respect to the order in which the data cube is arranged. The option `indexing='ij'` is needed to get the right order, and it was missing in the code that produced the grid.

As a result, in every cell the correct velocity was projected onto the **wrong direction**, with x and y exchanged. The figure shows this with illustrative values: for a cell at $(3, 1)$ with velocity $\mathbf{v} = (300, -200)$ km/s, the correct projection gives $v_r = +221$ km/s (the cell is receding); using the direction of $(1, 3)$ gives -95 km/s, and the same cell appears to approach.

CF4++: why the radial velocity grid was wrong

Radial velocity is the velocity $\mathbf{v}$ projected onto the cell direction $\hat{r}$: $v_r = \mathbf{v} \cdot \hat{r}$



The cell velocity v is correct: it is the direction it is projected onto that is wrong (x and y swapped).

The vx, vy, vz and density grids are correct; only vr_mean_CF4pp and vr_std_CF4pp are affected.

Check in HALO (cf4_grid.py): published vr vs projection of v — median difference 115 km/s; with x and y swapped: 0.00 km/s.

Error confirmed by A. Hollinger (ANU); corrected file being released by the CF4++ team (H. Courtois).

Cell position and v are illustrative values.

What is correct and what is not

- The **density** grid and the **Cartesian velocity** grids v_x , v_y , v_z are correct.
- Only the two derived grids are wrong: **vr_mean_CF4pp** (mean value) and **vr_std_CF4pp** (uncertainty), because both depend on the same projection.
- HALO's flow diagrams and basin-of-attraction maps are not affected, because they use v_x , v_y , v_z and not v_r .

How HALO noticed it

The module `cf4_grid.py` runs a consistency check: it recomputes $\mathbf{v}_r$ by projecting the Cartesian velocities and compares it with the published $\mathbf{v}_r$ grid. The median difference was **115 km/s**, far too large to be noise. Swapping x and y in the unit vector $\mathbf{r}$, the median difference dropped to **0.00 km/s**: the published grid matched exactly a projection made with the axes exchanged.

The confirmation

- 27 September 2026: report sent to H  l  ne Courtois (Universit   Claude Bernard Lyon 1) and Amber Hollinger.
- 30 September: Amber Hollinger (now at ANU) confirms the cause, the missing `indexing='ij'` in `np.meshgrid`.
- 9 October: confirmation that **`vr_std_CF4pp`** is affected too. The corrected file will be uploaded by Courtois with a note asking users to replace the previous version. Explicit permission to mention the finding publicly.

The team rarely uses the ready-made radial grid and, when needed, recomputes it from the Cartesian velocities: this is why the error never showed up in their own work.

The message for the audience

You do not need a large telescope to contribute to research. It is enough not to trust data blindly, even when published by a leading team, and to check it independently. This is exactly HALO's principle: every quantity is verified, nothing is taken for granted.